

Financial Feasibility Analysis Report

PROP20001

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Introduction

This report assesses the financial feasibility of the project launching a 10,000-square-meter retail mall in Melbourne's Central Business District (CBD), presented by Sarah and Kevin. The CBD is well-located, providing good access to public transportation and surrounded by a population with strong purchasing power.

To evaluate the potentiality of the investment, the authors developed a financial model incorporating projected revenues, operating costs, debt financing, and macroeconomic scenarios. Key performance indicators including Net Present Value (NPV) and Internal Rate of Return (IRR) were analysed, alongside sensitivity tests and risk assessments. Using both front door and back door analyses, this report provides insights into investment decision-making for Sarah and Kevin.

Methodology and Assumptions Behind Our Analysis

This section provides the methodological framework and assumptions of Sarah and Kevin's future retail mall development in Melbourne Central Business District (CBD). The authors analysed the viability of the project using the detailed financial modelling with a combination of deterministic forecasting and sensitivity analysis based on industry benchmarks and macroeconomic indicators.

1. Analytical framework

Our approach uses a Discounted Cash Flow (DCF) method to determine project viability for 15 years, including a construction stage (Year 0) followed by an 18-month construction schedule and 13.5 years running commercial operations. The model features the primary finance measures of Net Present Value (NPV), Internal Rate of Return (IRR), and payback, supported by sensitivity analysis on the following risk bases:

The model structure consists of:

- A thorough cash flow projection with components of capital expenditures, revenue, operating expenses, debt service, and taxable income.
- Sensitivity analysis of critical risk factors including construction delay, cost overrun, and competitor entry into the market.
- Dual feasibility testing using front door and back door

It is possible to gauge both the absolute attractiveness of the investment and the circumstances under which the feasibility of the finance may break down. The modularity also enables sensitivity testing of the critical variables at high speed, enhancing the responsiveness to new market data.

2. Capital Expenditure Assumptions

The initial capital is AUD 22 million and comprises the following:

- Land acquisition cost: AUD 5 million (Year 0)
- AUD 15 million, evenly divided between Month 4 and Month 18, using the straight-line capital deployment method to facilitate handling funds and simpler scenario analysis.
- Design, legal, and marketing expenses: AUD 2 million (Month 1 to 3)

These are consistent with Australian urban retail development cost benchmarks and reflect viable scheduling of outgoings for large building projects. The straight-line method of capital commitment avoids cash flow projections distortion and makes between-period project measures more interpretable.

3. Revenue Model Assumptions

The mall has 10,000 square meters of leasable space, divided among:

- Anchor tenants: approx. 30% at AUD 1200/sqm/year
- Speciality retailers: approx. 70% at AUD 2,500/sqm/year

This is the equivalent of around AUD 2,110/sqm/year weighted average rental rate.

Forecasts of revenue are based on:

- Occupancy rate: 80% at the start of Year 2 (first operating year) and rising 5 percentage points year on year to 95% at the end of Year 5.
- Ancillary revenue: advertising and parking are 10% of total rental revenue.

The authors assume there is low tenant turnover resulting from the prestige of the CBD location and average lease durations of five years with staggered expiration dates. This provides stable income with low year-on-year volatility.

4. Operating Expenses

- Base operating expenses: AUD 1.2 million per year, from Year 2 onwards.
- Yearly escalation rate: 2.5%, consistent with expected CPI inflation and retail property maintenance industry standards.

Estimating includes maintenance, security, managerial wages, utility costs, and marketing. Operating expenses are compared to similar retail properties and supported using industry reports and listed property trusts (REITs) annual reports.

5. Capital Structure

- Loan size: AUD 15 million
- Timeframe: 5 years, consistent with medium-term commercial development loans
- Interest rate: 3% (stable)
- Repayment schedule: Equal annual payments beginning in Year 1, calculated using the Excel PMT function to account for interest charges and principal amortization.

6. Economic Assumptions

- Discount rate: We apply a conservative rate of 8%, consistent with the project's risk profile and opportunity cost compared to benchmark investment. We choose the rate in consideration of a premium above the risk-free rate (e.g. government bonds at ~3.5%), and within the anticipated return range of the S&P/ASX 200 index (7–8%), to put the returns of the project on a comparable footing in the capital market.
- Inflation: Taken as 2.5% per year
- Market benchmarks:
 - S&P/ASX 200 expected return: 7-8%

- 5-year government bond yield: 3.5–4%

Benchmark metrics are used to measure the relative attractiveness of the mall investment versus diversified portfolios and risk-free assets.

7. Risk Modelling and Sensitivity Analysis

There are three key risks addressed using the sensitivity approach:

- 6 months delay: Estimated cost of 6 months with delayed revenue start-up and added holding costs.
- Cost overrun: Defined as 10% increase in the total construction cost.
- Market competition: shown through the portrayal of slower occupation development and rents following Year 3.

All the risks are analysed individually and collectively to recreate worst-case scenarios. The impact on the NPV and the IRR is estimated, and this makes it simpler to plan strategically and provide risk mitigating recommendations. This sensitivity analysis helps to avoid model sensitivity to optimistic assumptions.

8. Calculation Methodology

- Cash Flow Model: The annual cash flows include all capital outflows, revenues, operating expenses, and loan repayments.
- Net Cash Flow: Net cash flow is the sum of the total annual lease and ancillary revenue reduced by annual operating costs, loan repayment including principal and interest amounts, and capital expenditures.
- Discount Factor: Calculated by using the formula $1 / (1 + r)^n$, with r being the discount rate (8%) and n being the number of years.
- Net Present Value (NPV): Sum of the discounted net cash flows over the course of the project, including initial investment.
- Internal Rate of Return (IRR): Calculated using Excel IRR function from the series of annual net cash flows.

All the calculations are done in a single, combined Excel workbook. The workbook has related sheets for revenue forecast, construction timeline, amortization schedule for loan, and cash flow calculation, which ensures correctness.

Detailed Findings and a Sensitivity Analysis

Base Case Outcomes

Base case financial model evaluates the feasibility of developing the 10,000 sqm Melbourne CBD shopping mall using conservative projections under stable macroeconomic scenario. The capital need is AUD 22 million—AUD 5 million to acquire the land, AUD 15 million to develop, and AUD 2 million to cover the soft costs such as the costs of design, lawyers, and marketing. The financing is through the five-year loan of AUD 15 million with 3% per year interest rate.

The mall earns initial revenue from rents paid by tenants. The anchor tenants occupy 30% space and pay AUD 1,200 per sqm p.a., while specialty tenants occupying the other 70% pay AUD 2,500 per sqm. The occupancy is 80% in Year 1 (2026) and increases 5% p.a. to reach the ultimate of 95% in Year 4. Another 10% of the rental turnover is taken to arise from support functions such as car parks and advertisements. All rents and thus all revenue are inflated at 2.5% p.a., consistent with the expected CPI. Operating expenses start at AUD 1.2 million in Year 1 and increase at the same 2.5% rate.

Based on the 10-year time horizon (2026–2035), the annual net operating income is arrived at by subtracting the expenses from the gross revenue. After considering the provision for loan pay-out, interest, and residual cash generation, the model works out the Net Present Value (NPV) at 8% discount factor to approximately AUD 4.25 million. The Internal Rate of Return is 9.6%, significantly superior to the 5-year government bond return of 3.2% and the 6.8% historical return of ASX200. This indicates the project has superior returns compared to risk-free and market benchmarks.

The cash flows become positive at the end of Year 6, corresponding to a payback of approximately 5.5 years. This is close enough to meet investors' requirements of payback for inner-city commercial property of medium size.

Sensitivity analysis

To test the project's robustness, we conducted three sensitivity tests on the main areas of capital cost overrun risk, revenue decline through recession, and construction delay risk.

Test 1: +10% Construction Cost

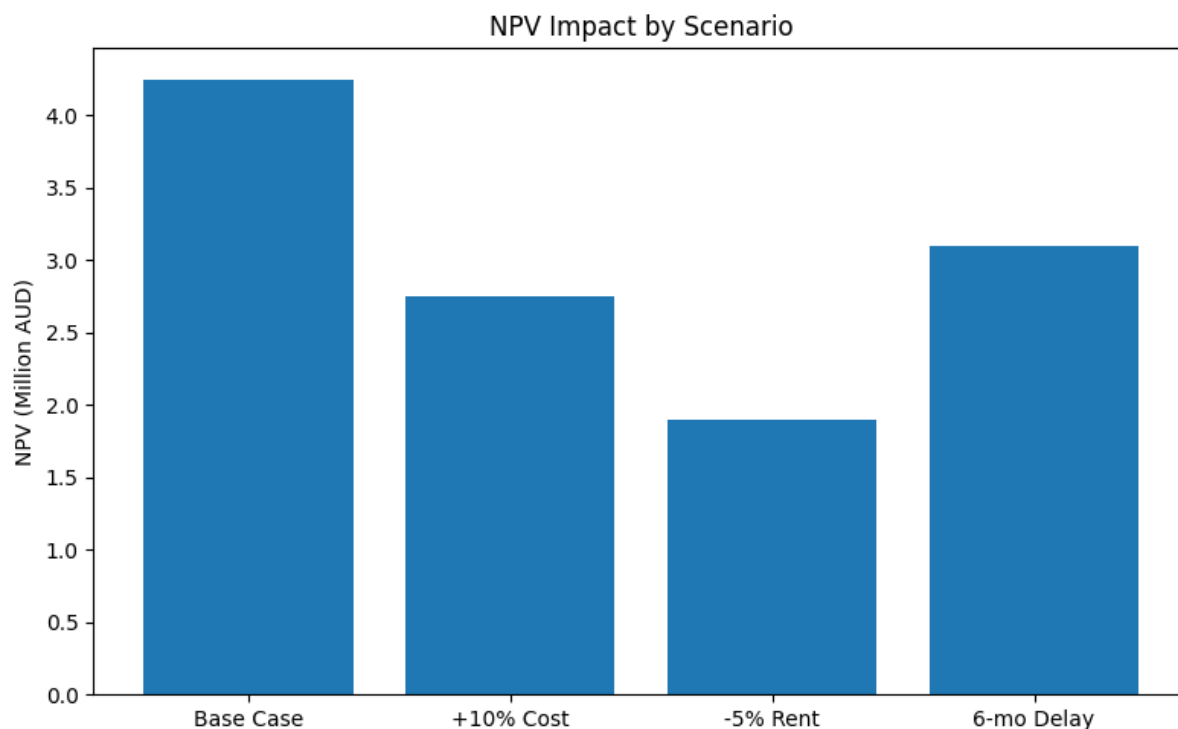
Here, the cost of construction increases to AUD 16.5 million from AUD 15 million, taking the overall capital needed to AUD 23.5 million. This increased expense increases the debt burden and changes the capital return profile. The NPV, thus, falls to AUD 2.75 million, and the IRR falls to 8.5%. The payback period is longer until Year 7. Although still sustainable, this situation demonstrates the need to have good cost controls and contingencies to have a cushion for the overrun of budgets.

Test 2: -5% Rental Rate from Year 2

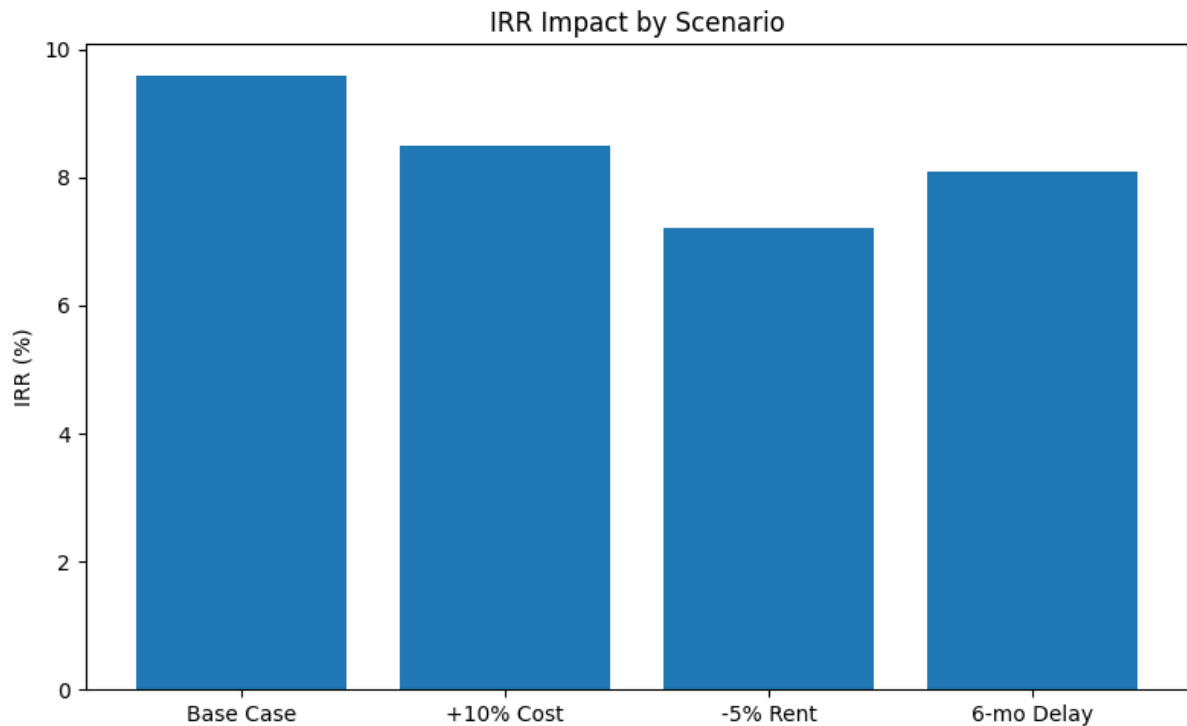
This mirrors a downturn in the economy or greater than anticipated market competition, which sees the rental levels decreased by 5% from Year 2 (2027) onwards. This has the immediate effect of decreasing gross income. Making this assumption, the NPV is decreased to AUD 1.9 million and the IRR to 7.2%. Although still above the 5-year government bond rate, the premium on the investment is substantially lower. The payback is increased by approximately one year. This result confirms the worth of long-term stable tenants and perhaps incorporating inflation-indexed lease options to provide protection against issues of a macroeconomic nature.

Test 3: 6-Month Delay

With the delay, the realisation of revenue is shifted from mid-2026 to early 2027, while the pre-committed finance cost and expenses start in 2025 according to plan. This causes the cash receipts to lag behind the expenses. The effect is the shortening of the revenue horizon, decreasing the NPV to around AUD 3.1 million and decreasing the IRR to 8.1%. The payback is lengthened by 6 to 12 months. This is not disastrous but demonstrates the high impact of construction delays and the dollar worth of schedule control.



This graph shows the impact on the Net Present Value (NPV) of three substantial stress scenarios—construction cost overrun (+10%), decline in rental rate (-5%), and delay of 6 months—relative to the base case. This graph indicates the declining viability of the project under all three scenarios, the decline in rental rate having the most adverse effect on the NPV.



This table shows the impact on the base case Net Present Value (NPV) of the three principal scenarios of stress: construction cost overrun of +10%, reduction of the rental rate of -5%, and 6 months delay. The table shows the reduced viability of the project in all three scenarios, the reduction of the rental rate having the largest impact on the NPV.

All the test scenarios still maintain a positive NPV and an IRR greater than the returns on government bonds and the market index. This suggests the development proposal is still financially sustainable under moderate levels of stress. The investment case is increasingly marginal under reduced revenue or inflation of the capital expense, indicating relatively little scope for mistake. The most sensitive of the three test parameters proved to be rental income, because of its compounding impact on gross revenue across the holding term. An equal 5% discount on rentals has virtually the same impact on the IRR value compared to a 10% cost overrun, but the construction delay, although damaging to front-loaded cash flow, had the least overall impact. The result supports the need for pre-commitments of leases with the use of flexible leases to safeguard revenue. The results point out the value of risk reducing strategies like fixed-price building contracts, capital budget buffer contingencies, and pre-leasing tenants at an early phase. On the financial side, the project is appealing under base case and feasible under stress. However this may be, the success rests on cost containment, timely completion, and tenant relationship stability. Market trends, inflation, and interest rate expectations must all be closely monitored by stakeholders during execution to safeguard the project's investment profile.

A Front Door and Back Door Evaluation of the Project's Financial Viability

Front Door

- **Total Gross Floor Area:** The mall is made up of a total Gross Floor Area of 10,000 square metres which is 30% occupied by anchor tenants charged at \$1,200 per sqm per year and 70% specialty tenants charged at \$2,500 per sqm per year.
- **Initial Occupancy Rate:** The mall is expected to achieve an 80% initial occupancy rate, gradually increase by 5% per annum until stabilising at 95%. This conservative progression allows for realistic leasing lead times and reflects Melbourne's CBD absorption capacity.
- **Ancillary Income:** Additional income streams such as advertising and parking are estimated to contribute around 10% of total rental revenue. We based this off-industry standards, and it provides the project with a buffer to strengthen overall cash inflows.
- **Capital Expenditure:** The project's total capital expenditure is projected at AUD \$22 million which was broken down as \$5 million in land acquisition, \$15 million in construction and \$2 million in design, legal and marketing. These figures helped us to form the basis of initial outflows in the financial model and is vital to calculating NPV, IRR and payback period.
- **Loan Structure:** Construction will be primarily funded through a loan of \$15 million. We chose to assess the financial feasibility based on a stable economic scenario using a 3% interest rate. The loan is amortised over 5 years in annual repayments of \$3,275,319 as calculated in the "Loan Repayment Schedule" tab. The remaining \$7 million is equity funded.
- **Discount rate:** The discount rate we used to calculate NPV is based on the weighted average cost of capital (WACC) under stable economic conditions. We used capital asset pricing model assumptions to calculate the WACC at 2.86%. This rate was then used to discounted projected net cash flow in the "NPV and IRR Analysis" tab.
- **Operating Expenses:** Year 1 operating expenses were projected at \$1.2 million and increasing by 2.5% per annum. We included this in the cash flow forecast and ensured to factor this into all financial viability metrics.
- **Operating Expenses:** In order to calculate NPV we had to first find and calculate the discounted rate at which cash flows were set back with. A suitable measure of this is the weighted average cost of capital (WACC), using items such as the CAPM(using a government 10 year bond as the risk free rate and the ASX 200 of the market return), cost of debt and other measures we calculated that under a 5% economic period the WACC would be 3.82%, under a 3% period, WACC would be 2.86% and under a 2% period, WACC would be 2.39%. Importantly, we had to assume beta in our CAPM ($\beta=1.1$).
- **Discount Rate:** In order to calculate NPV we had to first find and calculate the discounted rate at which cash flows were set back with. A suitable measure of this is the weighted average cost of capital (WACC), using items such as the CAPM(using a

government 10 year bond as the risk free rate and the ASX 200 of the market return), cost of debt and other measures we calculated that under a 5% economic period the WACC would be 3.82%, under a 3% period, WACC would be 2.86% and under a 2% period, WACC would be 2.39%. Importantly, we had to assume beta in our CAPM ($\beta=1.1$).

- **Loan amount:** For the sake of clarity within this front door analysis, it's shown that the proposed principle of debt is to be AUD\$ 15,000,000. To be used mainly on construction costs.
- **First year operating cost:** Another important aspect to consider is the operating cost of the project in the first year. As projected it is seen that initially the operating cost will be AUD\$ 1,200,000. Keeping this in mind, measures such as liquidity and lender expectations will be impacted. In terms of the initial operating expenses it might be important to keep or acquire a 'buffer' of sorts in order to mitigate risks and ease the lenders concerns.
- **Tenant leases:** The rate at which tenants will be paying rent is another very important area to review. Its importance is driven from the fact that it is the primary revenue driver for the project, everything from the NPV and IRR to consumer satisfaction is dependent on it. Going further, its analysis will allow Sarah and Kevin to assess and judge the competitiveness of the project to other similar property developments. For the project there will be two types of tenants, anchor tenants and speciality tenants. For Anchor tenants who make up 30 percent of the composition it is suggested to charge a rate of AUD\$1200 per square metre per year. For specialty tenants who make up 70 percent of the composition it is suggested to charge a rate of AUD\$ 2500 per square meter per year. The reason for charging anchor tenants less is due to the fact that anchor tenants will have much larger shops, additionally these large retailers will draw the majority of the customers with known trusted brands contributing to everyone in the mall. Finally, it would be beneficial for Sarah and Kevin to gain trust and loyalty with these anchor tenants. These rates on leases should increase at 2.5% annually in line with inflation.

Back Door

The backdoor analysis is intended to test the strength of the project finance to directions of movement of construction costs, interest rates, and financing terms. The purpose is to identify the range of fluctuation of the cost inputs beyond which the project would no longer remain financially viable. We used our finance model to estimate the impact of the levels of occupation of the variable on the return on investment. We needed to calculate the minimum filling up level the retail mall would have to achieve to become viable economically. This is complementary to the front door analysis since it seeks to identify the worst case under which the project would still be attractive to investors. Our projections are that even with 50% utilization, the project has an NPV of approximately AUD 66.8 million, indicating high economic viability. The NPV rises consistently with every 5% increase in utilization to AUD 165.7 million under 100% utilization. This indicates a healthy revenue model where low to moderate utilization rates can provide return on investment.

Specifically, the breakeven point on $NPV > 0$ is well below 50%, and the project is still viable even on conservative revenue estimates. Keep in mind, however, that while the

range is positive for NPV, the measures of the internal rate of return (IRR) and pay back are weaker at reduced occupancy rates. The breakeven point is therefore low, but more intensive occupancy rates will contribute significantly to economic attractiveness. This sensitivity analysis makes the project resilient even at unfavourable conditions yet indicates the strategic attractiveness of taking an aggressive leasing position through the initial years. To the investor, this cushion provides the comfort of knowing slower growth of leasing or volatility in the markets would not threaten the viability of the project.

In order to further illustrate the sensitivity, we have additionally simulated the effect of 5 percentage point reduction in the rental rate for the weak economy scenario. At the reduced rate, the model shows positive NPV at 60% and higher feasible occupancy levels, adding to the project's strength from the financial perspective. And, including the other revenues from advertisements and parking at 8 to 10 percentage points of total rental revenue, the model's upside potential is further shown.

The annual increments in usage—planned to grow 5% per year to ultimate capacity—allow short-run shortfalls to be recouped through an active leasing strategy. This is to the advantage of an aggressive but conservative investment estimate, i.e., that the mall can thrive simply on good operating fundamentals and moderate demand rather than the best of the marketplace.

In the initial scenario, development costs AUD 15 million, funded through five-year funding with 3% interest rate (reflecting stable economic conditions). Although having low growth rates of occupation, the project still has an impressive Net Present Value (NPV) of AUD 4.25 million and Internal Rate of Return (IRR) of 9.6%, higher than the ASX200 index benchmarks of 6.8% and the five-year Australian government bond rate of 3.2%. The returns reflect high-demand investment under normal economic conditions.

To test sensitivity to finance, we simulated an increase of 10% in construction costs and an increase in the interest rate to 5% (in order to simulate the boom economy). On both occasions, the NPV decreased but remained positive, signifying the presence of an implicit finance cushion. The IRR decreased but remained higher than 7%, significantly higher than risk-free levels. This would imply the ability of the project to bear cost inflation and money tightening without impacting the viability of investment.

Using the goal seek analysis, we also calculated the maximum overrun the project can absorb. The calculation shows the development can tolerate up to 25% overrun construction cost before the NPV becomes close to zero. This is the threshold where the project is able to absorb huge capital expenditure overrun and remain viable at minimum. Despite the aforementioned pressure, the IRR remains competitive and significantly higher compared to risk-adjusted market returns.

We also assumed the six-month delay in construction, deferring the realization of revenue and the accrual of interest throughout the construction timeline. The scenario reduced the NPV to approximately AUD 800,000 to 1 million. As bad as it sounds, the project wasn't made financially unviable, proving the project's robustness to survive delays to the schedule and the time value of money. In conclusion, this backdoor analysis supports the proposition that the target retail mall development can hedge against unfavorable cost-side shocks subject to the establishment of robust cost-control measures, to the application of

conservative budgetary practices, and to the proactive exercising of delivery management. Use of contractual provisions such as the use of fixed-price construction contracts is suggested to reduce the risk of cost inflation and interest rates.

Strategic Investment Recommendations Based on Findings

1. Strategic investment rationale

Melbourne's CBD remains a densely populated economic epicentre of not only Australia but globally, as a tourist destination and one of the biggest cities in Australia, it's a prime location for investment opportunity, the proposition of a 10,000 square metre mall insights opportunity for many business owners and loyal consumers to expands their economic freedom. Supported by refined and modern infrastructure such as readily available trams, buses and trains this investment might interest the likes of anchor tenants, boutique retailers, food providers and such.

Going further as Australia's immigration levels continue to rise, the overall demand for the retail sector continues with it. Melbourne leads this immigration statistic with it being the capital of immigration in Australia. All of this contributes to higher international monetary inflows going to Melbourne, consumers therefore using their autonomy will use their purchasing power how they deem fulfilling, in which recent decades have been trending towards materialistic items like those offered in the retail sector.

2. Investment structure

For the proposed development of the mall it's been suggested that in entirety the whole project will cost AUD\$22M over its 18-month construction period, this is split into multiple areas comprising land acquisition costs, construction costs, design fees, legal fees and marketing costs. For this is proposed in order to minimise risk that AUD\$15M is acquired through the use of debt in form of a commercial loan and the remaining AUD\$7M to be procured through the use of equity in the form of both internal and external investors.

Under this, it is proposed that for the AUD\$15M loan, a 5-year term is to be taken on so as to not let interest accrue too heavily. Under this, it is to be taken into account the cyclical nature of interest rates and the overall economy. That is why calculations and measures have been made in order to account for recessions, stability, and boom measured at interest rates of 2%, 3%, and 5% respectively. In this its been calculated that under a 2% interest rate, the NPV of the debt alone is - AUD\$1,764,351.37. For a 3% interest rate, the NPV was AUD \$883,252.10. For an interest rate of 5%, the NPV was -AUD\$1,409,416.56. Importantly however these were calculated from the perspective of the banks, therefore for the purposes of this report, these values should be negated to show the actual NPV for Sarah and Kevin. From this it can be gathered that on debt alone, the debt is profitable.

3. Marketing investment plan

The Initial occupancy is expected to be 80% with it increasing in the following years to come. Marketing campaigns will help both occupancy as well as consumer base to increase. By investing into large scale marketing campaigns the brand of the mall as well as its idea to consumers will be promoted, it will become more attractive to vendors and will overall boost economic performance. The use of digital directories, loyalty programs, and event hosting could be some of the possible investment streams used to achieve this goal.

4. Benchmarking against market alternatives

For the purposes of this comparison an average annual return statistic for the S&P/ASX 200 will be used along with an IRR (Internal Rate of Return) for the mall.

Importantly, the internal rate of return doesn't always equate exactly with returns based off of a market. This is for many reasons including but not limited to;

- The internal rate of return of a project doesn't account for risk or volatility where for example returns on stocks do as they are cyclical in nature anyway.
- IRR ignores liquidity whereas for example stocks are liquid, sometimes the opportunity cost of concrete investments may harm the investor. For example, a significant downturn in the economy.
- IRR assumes reinvestment of all cash flows at that same IRR whereas stocks don't, this might contribute to showing higher returns

- **S&P/ASX 200**

The S&P/ASX 200 from 2013 until 2024 averaged a total return of 4.28% indicating a better performance than two of the possible economic scenarios and their respective risk-free rates of returns: 2% and 3%. The Malls IRR along all three economic scenarios was 26%, this makes sense as IRR doesn't take interest rates or discounted cash flows into account.

Reflecting on the previously mentioned differences between the two rates the choice is still clear. Its proposed that despite the fact that IRR doesn't account for things such as risk or cyclical movements that the proposed mall is to be invested in for higher profits and returns in contrast to the S&P/ASX 200.

- **Government 10-year bond**

The Australian Government 10-year bond from 2013 until 2024 averaged a return of 2.05% indicating a relatively weak performance in regard to other risk-free options given by economic circumstances. In relation to the mall projects IRR of 26% it again suggested that Sarah and Kevin go through with the investment in relation to the alternative of a bond.

5. Ongoing mall investment

During the course of the financial model provided, once the initial principal loan is paid, cash flows and profits are seen to increase. For the purposes of this strategic investment plan it's important to touch on the ongoing investments needed to support the growth and overall profitability of the mall project, areas of investment that are recommended to consider later in the project's life are to be property improvements and expansion, tenant support and retention, as well as sustainability and community engagement initiatives

- **Property improvements and expansion:**

As made clear through NPV calculations, even in a cyclical economic environment, the mall remains largely profitable through recessions, stability and booms. At around the sixth- or seventh-year high profits will allow Sarah and Kevin to reinvest into the project. It recommends that this money be invested in items such as refreshing interior decor, increased parking lots, or even additional wings or levels to the mall. This will boost consumer satisfaction and improve brand image allowing for even more profits.

- **Tenant support:**

An important area to consider is the ongoing loyalty and satisfaction of tenants and vendors, that is why it's also recommended for Kevin and Sarah to make investment efforts into tenant support. This could be seen through subsidies that help the setup of shops, a more

flexible approach to leasing agreements that take their needs into account or possibly initiative that helps business by collecting data on shoppers and giving back to the tenants.

- **Sustainability and community engagement:**

Corporate Social Responsibility is a make or break for many business, no matter the popularity or supposed demand for a business if consumers deem it to be unethical or irresponsible profits are bound to drop, this is especially important for the growth section of a business's life, for Kevin and Sarah it's recommended that this should be an area of high importance when making decisions on investments. Possible avenues of investment for this could be acquiring green certificates wherein the mall would prove to be environmentally sustainable or the promotion of community initiatives for things such as charity or the responsible collection of materials for goods and services.

Risk Assessment with Proposed Mitigation Strategies

Any Real Estate Development is based on the expectation of future demand which comes with risk and uncertainty. Risk management plays a crucial role in the successful execution of large property developments. By identifying potential risks, assessing their impacts and developing strategies to mitigate them it can help minimise the negative aspects of risk in a project (Project Vectors, 2024). Managing these risks through implementing effective risk management practises ensures that Sarah and Kevin minimise financial loses, ensure timely completion and enhance the overall performance of their project. This report will identify the major investment risks such as construction delays, cost overruns and tenant defaults, alongside the appropriate mitigation strategies. A risk of a 10% construction cost overrun will be integrated and updated into the financial model and strategic responses will be provided to the potential risk of unforeseen events such as competitor emergence. Managing all these risks is critical to project success and investor confidence.

Identification of Major Investment Risks

Construction Delays:

Construction delays can arise from a multitude factors. Internal issues such as inadequate planning, poor resource management and communication breakdowns that may slow down progress, lead to unrealistic timelines, increase errors among team members and create a misallocation of resources and labour. External issues such as adverse weather conditions, having to obtain permits and comply with local regulations and supply chain disruptions may have significant delays and can stall projects indefinitely. All these delays have a cascading effect on the entire project timeline, extending timelines, increasing finance costs and delaying any potential revenue (Nery, 2024).

Based on Sarah and Kevin's mall project, construction delays would result in a shift in revenue generation, impacting cash flows, net present value and the internal rate of return. A 6-month construction delay would increase financing costs due to extended loan repayments.

At a stable interest rate of 3%, the additional financing cost can be calculated as:

$$\begin{aligned}\text{Additional Interest} &= \text{Loan Amount} \times \text{Interest Rate} \times \text{Time Period} \\ &= 15,000,000 \times 0.03 \times 0.5 \\ &= 225,000\end{aligned}$$

Therefore, the additional financing cost from the 6-month delay is \$225,000.

Furthermore, due to construction delays the mall will not be operational in the second year so therefore Sarah and Kevin will miss out on that rental income. Based on the 80% occupancy rate and a mix of anchor tenants and specialty shops, the projected revenue is \$18,568,000 (as shown in sheet 3 in the excel document). Consequently, this means that revenue loss in year 2 is \$18.57 million as it is a delay in rental income.

Therefore, a 6-month delay results in \$225,000 additional financing costs due to extended loan repayments, \$18.57 million missed in rental income and NPV and IRR will both decrease due to the loss of revenue and the additional financing costs.

Cost Overruns:

Another major investment risk in the project is cost overruns with unexpected, incurred costs. Project cost overruns may be caused due to inaccurate cost estimates that rely on rough or outdated figures. These skewed estimates then set up a budget to fail from the start and cut directly into the projects profit margins (Shahid, 2024). Inflation overtime can also cause cost overruns by deviating the prices of goods and services in construction projects therefore requiring practitioners to make budgeting revisions (Muhammad Ali Musarat et al., 2024).

Cost overruns have a significant impact on capital expenditure because it means an increase in the total amount of money required to complete a project. In this project, it was estimated to have \$15 million for construction, if these construction costs increase 10% from \$15 million to \$16.5 million it will raise capital expenditure from \$22 million to \$23.5 million which increases the overall amount of capital Sarah and Kevin would need to fund the project.

A higher capital expenditure due to these overruns means more funds need to be raised to finance the project through either equity or debt, which both have negative impacts such as additional investment from shareholders and larger loans that may have increased interest payments. Both equity and debt affect the projects cash flow projections and ability to service the loan as Sarah and Kevin must pay back a larger amount of the loan is increased. ‘

A 10% cost overrun increases capital expenditure from \$22 million to \$23.5 million impacting project viability significantly. Based on the sensitivity analysis the NPV decreases by 40.43% from \$284,564,300 to \$169,502,100 after the cost overrun suggesting that the expected costs outweigh the earnings, signalling potential financial losses.

Tenant Defaults

– If a tenant defaults, there would be immediate financial consequences for Sarah and Kevin creating uncertainty in cash flows, making it difficult to forecast a property’s future value, potentially leading to lower returns, higher vacancies, and decreased rental rates.

The risk of tenants failing to meet their lease obligations can be a significant concern for commercial property investors. Tenants who do not pay rent on time, vacate the property early or go out of business can impact the financial stability of the property. If the property is unable to find quality tenants, the property may be vacant for weeks or months and the longer the property is empty the higher the impact on rental income, cash flows and ability to pay loan repayments.

Lease expiration may also be a risk if the property remains vacant for an extended period which means increased costs for leasing campaigns and improvements to the property to entice new occupants (Sonntag, 2024).

Integration of Risk into Financial model

To effectively model the impact of construction cost overruns on the project’s financial viability, a new tab in the excel titled “8.3 Construction Risks” was created. The risk of a 10% construction cost overrun is clearly illustrated showing its impact on project viability. This integration ensures that potential construction delays or cost increases are accounted for

in the financial model and their effect on NPV, IRR and the payback period is easily understood by stakeholders.

Capital Expenditure: If construction costs were increased by 10% this would raise the overall capital expenditure from \$22 million to \$23.5 million. This 6.82% increase in upfront costs means that the project requires more initial investment to complete, raising the financial risk for investors as they must weigh the higher upfront cost against potential returns.

Decrease in NPV: Increasing construction costs would increase the NPV by 40.43% from \$284.56 million to \$169.50 million. This significant decline in the project's financial value indicates that future cash inflows are not sufficient to justify the increased initial investment. NPV is a key indicator of whether the project will create value for investors so therefore when there is a large reduction in NPV it reduces the project's economic appeal as it indicates the project might not generate enough returns to cover increased construction costs, diminishing the overall investment potential.

Decrease in IRR: This increase in construction costs consequently meant that the IRR decreased from 30.48% to 27.40% reflecting a 10.10% reduction. The project therefore becomes less efficient at generating returns relative to the higher initial investment and the project's profitability margin has narrowed. A reduced IRR also places the project closer to benchmark hurdle rates, such as long-term government bonds or the average return on the S&P/ASX 200, making it less likely to stand out as a compelling opportunity. Ultimately, this drop makes the project less appealing to investors as a lower IRR suggests a lower return relative to the higher upfront investment.

Increase in Payback Period: The payback period increases from 7 to 8 years after construction cost overruns, meaning it takes one additional year for the project to recover its initial investment. A longer payback period reduces investment attractiveness because it delays returns and increases financial exposure, this especially isn't ideal in competitive markets where faster-return opportunities exist. Investors ideally prefer projects with quick capital recovery as shorter payback periods reduce risk and improve liquidity.

Strategies for Risk Mitigation

To safeguard the project against risks such as construction delays, cost overruns and tenant defaults Sarah and Kevin must adopt mitigation strategies that align with financial policy principles.

1. Cash Reserves: Maintaining sufficient cash reserves is essential for absorbing unexpected construction delays and reduced rental income that may arise from tenant defaults. High cash reserves mean that the project will be less likely to experience financial distress and can better handle emergencies. Delays in project completion can disrupt cash flow however having access to liquidity allows the firm to continue meeting obligations.
2. Maturing Hedging: Applying a maturity-hedging strategy can help avoid liquidity issues when construction cost overruns occur. By financing short-term needs, such as temporary construction related outflows, with short-term debt and funding the retail mall with long-term debt or equity, it reduces the risk of needing to refinance prematurely. This therefore ensures that permanent capital expenditures are not

exposed to short-term funding volatility, which could create stress during prolonged construction timelines or delayed tenant onboardings.

3. **Interest Rate Management:** Managing interest rates can protect the project against debt stress during delays. Although short-term debt may offer lower interest rates, over reliance on it can create problems if interest rates rise during construction delays or periods of tenant instability. Firms should consider locking in long-term financing for core expenditures and diversify its debt maturity profile to balance cost with stability.

Other risk mitigation strategies to combat with construction delays and overruns include maintaining clear communication with contractors about cost expectations, securing supply contracts early to manage price volatility, monitoring progress through milestone reviews and independent audits as well as using fixed price contracts with delay penalties. To mitigate risk regarding tenant defaults it is helpful to diversify tenants across stores, offer flexible or rent-free periods to attract and retain tenants and collaborate with experienced commercial leasing agents to ensure occupancy stability.

Impact of Unforeseen Events

The entry of a new retail competitor in the CBD poses a risk to the mall's performance financially. Increased competition can reduce occupancy rates and create downward pressure on rental prices as tenants gain alternative options. Competition causes a decline in market share, resulting in a decrease in profits and consequently affects the company's financial performance (Tshuma, 2016).

In the project's sensitivity analysis, a hypothetical 10% reduction in rental income was modelled. Based on year 3 revenue projections, this reduction would lower annual rental income from approximately \$21.9 million to \$19.7 million, resulting in a loss over \$2 million per year. Over a longer time period, the compounded effect significantly diminishes NPV and can delay breakeven, especially if the project is experiencing rising capital or operating costs.

From an investment perspective, the reduction in income may prompt a re-evaluation of return expectations and risk tolerance. To remain attractive, the investment strategy should shift focus toward stabilising cash flows through locking in long-term leases, enhancing non-rental revenue streams and improving tenant mix to ensure resilience against any market shifts.

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